

## 5G/6G Communication Systems- ECA5604 (LAB)

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### **Experiment 4: OFDMA Resource Block Allocation**

**Aim:** To simulate and analyze OFDMA (Orthogonal Frequency Division Multiple Access)

resource block allocation and evaluate system performance in terms of resource utilization and user throughput using MATLAB.

#### **Objective 1:**

```
clc; clear; close all;
```

```
% System parameters
```

```
bandwidth = 10e6; % 10 MHz
```

```
rb_size = 180e3; % 180 kHz per RB
```

```
% Total Resource Blocks
```

```
total_RB = floor(bandwidth / rb_size);
```

```
% Users
```

```
users = 1:6;
```

```
% RB allocation (simple equal + remainder distribution)
```

```
base = floor(total_RB / length(users));
```

```
alloc = base * ones(1,length(users));
```

```
alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(users))) + 1;
```

```
utilization = (alloc / total_RB) * 100;
```

```
disp('Total Resource Blocks:')
```

```
disp(total_RB)
```

```
disp('RB Allocation per User:')
```

```
disp(alloc)
```

```
figure
```

```
% ----- Graph 1: RB Allocation per User -----
```

```

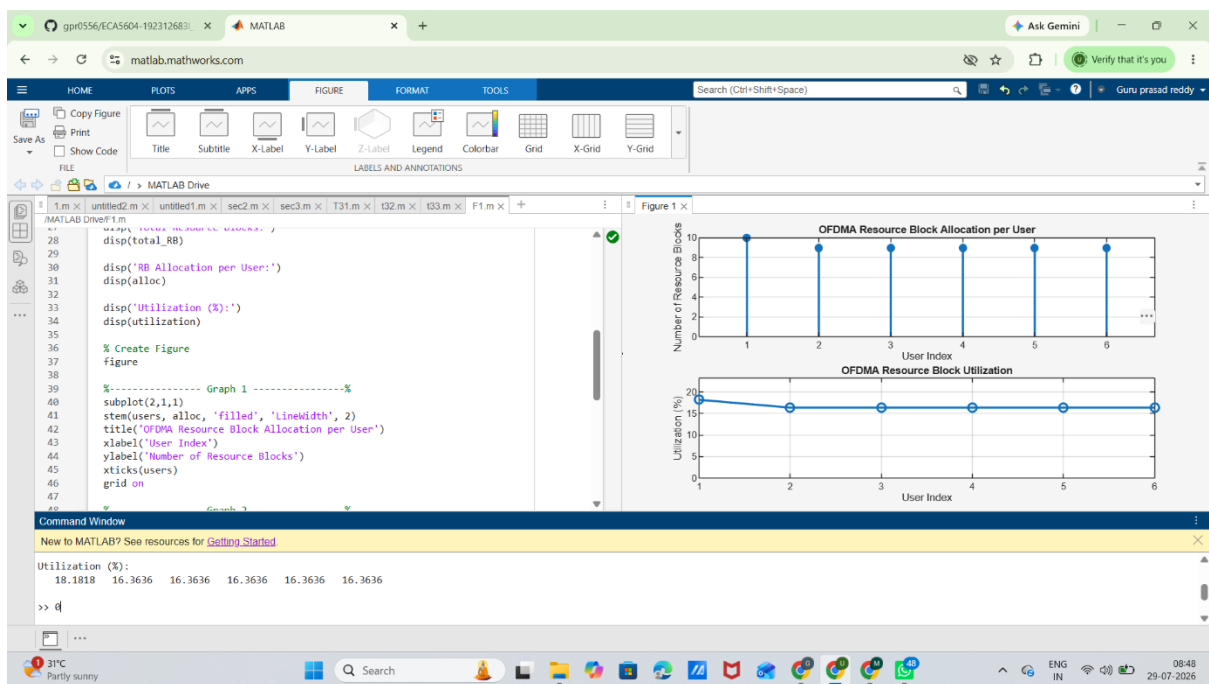
subplot(2,1,1)
stem(users, alloc, 'filled', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on

% ----- Graph 2: RB Utilization -----

subplot(2,1,2)
plot(users, utilization, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on

```

## Output:



## Objective 2:

### MATLAB Code:

```
clc; clear; close all;
```

```

% Total system resource blocks
total_RB = 50;

% Number of active users
users = 1:5;

% RB allocation to users (example scenario)
alloc = [12 10 9 8 11];

% Resource Block Utilization (%)
util = (alloc / total_RB) * 100;

disp('Resource Block Utilization (%) per User:')
disp(util)

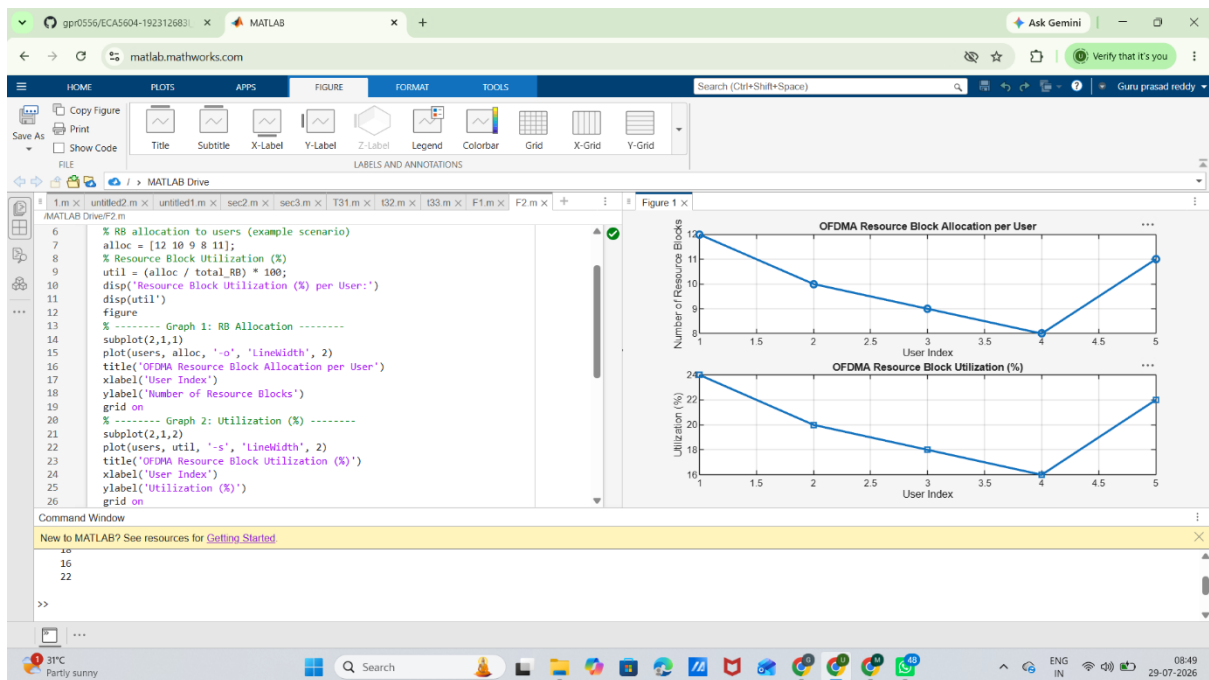
figure

% ----- Graph 1: RB Allocation -----
subplot(2,1,1)
plot(users, alloc, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on

% ----- Graph 2: Utilization (%) -----
subplot(2,1,2)
plot(users, util, '-s', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on

```

## Output:



## Objective 3:

### MATLAB Code:

```
clc; clear; close all;
```

```
% System parameters
```

```
rb_bw = 180e3; % Bandwidth per RB (180 kHz)
```

```
users = 1:5;
```

```
% Allocated Resource Blocks per user
```

```
alloc_RB = [12 10 9 8 11];
```

```
% Modulation efficiency (bits/symbol)
```

```
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
```

```
% Throughput calculation (Mbps)
```

```
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
```

```
disp('User Throughput (Mbps):')
```

```
disp(throughput)
```

```
figure
```

```
% ----- Graph 1: Throughput per User -----
```

```
subplot(2,1,1)
```

```

plot(users, throughput, '-o', 'LineWidth', 2)

title('OFDMA User Throughput')

xlabel('User Index')

ylabel('Throughput (Mbps)')

grid on

% ----- Graph 2: RB vs Throughput -----

subplot(2,1,2)

plot(alloc_RB, throughput, '-s', 'LineWidth', 2)

title('Resource Blocks vs Throughput Relationship')

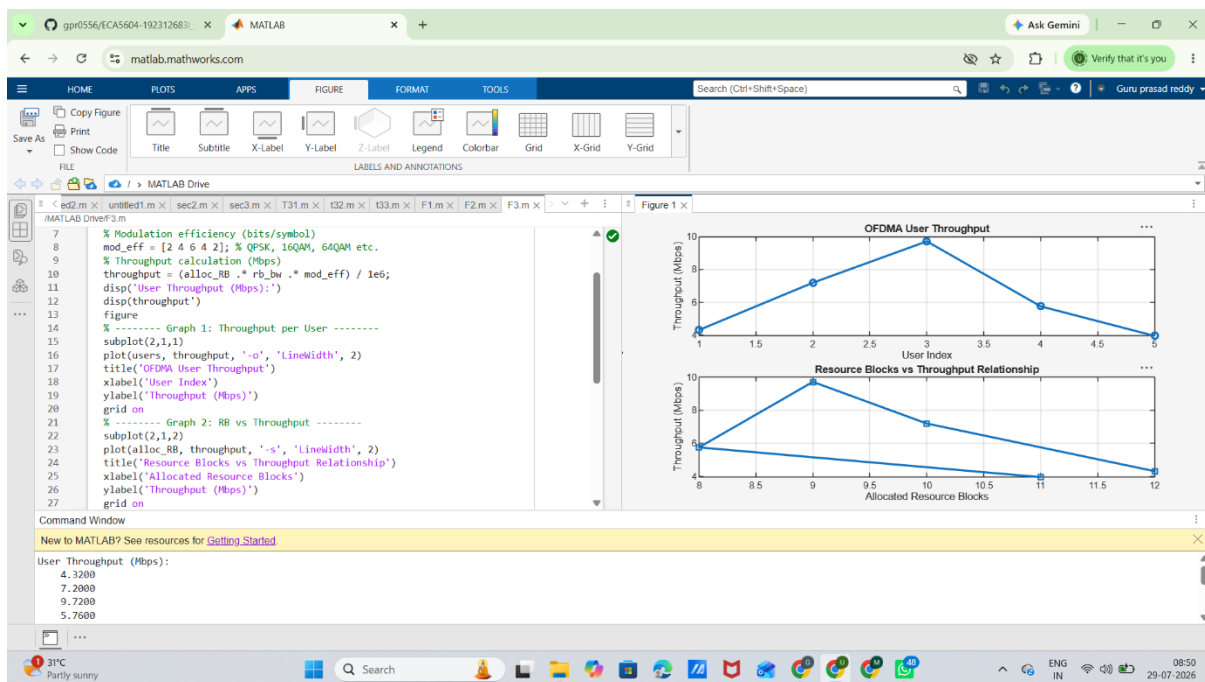
xlabel('Allocated Resource Blocks')

ylabel('Throughput (Mbps)')

grid on

```

## Output:



**Result:** The propagation loss for Sub-1 GHz, Mid-band, and mm Wave frequency bands was successfully analysed using MATLAB. The results showed that propagation loss increases with both frequency and communication distance, with mm Wave experiencing the highest attenuation and Sub-1 GHz providing the best coverage.